

PSC-Q (Addendum): Supplementary Evidence and Operator Microstudies

February 25, 2026

Abstract

This companion note preserves the PSC-Q microstudies and operator-facing “gotchas” that were previously shipped as a split supplement. The main PSC-Q paper focuses on the mechanism interface and declarations; this addendum collects short empirical illustrations and checklists. Figures are omitted in this source-only release; corresponding plots and raw tables are available on request.[3]

Scope. Addendum to PSC-Q.[1]

Imports. We import Congestion-EQ as the target contract and PSC-Q as the mechanism surface.[2, 1]

Exports. A compact set of sanity checks and microstudy patterns for implementers.

1 Supplementary evidence and microstudies (formerly a split supplement)

This appendix preserves the operator-facing microstudies and “gotchas” that were previously shipped as a separate PSC-Q supplement in the split archive. The unified archive keeps a single-file rule, so we inline the salient parts here. Figures are omitted in this source-only release; the corresponding plots and raw tables are available on request.

1.1 Additive padding can worsen congestion-TV in the high-utilization knee

The main text emphasizes *substitution* (idle-filling) because it can reduce congestion witnesses without increasing mean load. In contrast, *additive* padding can increase utilization and move the system closer to the slack “knee,” at which point small secret-correlated load differences yield large changes in waiting-time and RTT laws. Fig. 1 (from the research archive) illustrates a simple regime in which additive padding increases the measured congestion-TV.

1.2 Partial idle-filling is often enough

Even partial substitution (filling only a fraction of idle slots) can materially reduce congestion-TV without increasing mean load. Fig. 2 summarizes an archive sweep supporting this empirical claim.

Figure omitted in source-only release. Requested file: `fig_padding_worsens_tv.pdf`.

Figure 1: Archive microstudy: a “padding can worsen” regime when additive padding pushes utilization upward.

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Figure 2: Archive microstudy: partial idle-filling can already reduce congestion-TV without increasing mean load.

1.3 Substitution must avoid service bias (engineering pitfall)

Substitution is only “free” if cover is confined to otherwise-idle service opportunities. If cover work competes with real work (shared queue or shared downstream resource), it can bias service and increase delay. Fig. 3 illustrates this pitfall under a simple mixed-queue implementation.

Implementation checklist (non-exhaustive).

- Define at most one service opportunity per slot; do not create extra opportunities when real work is present.
- Treat cover as cancelable/preemptible: if real work arrives, cancel or defer cover so it never delays real work.
- If a separate queue is used, enforce strict priority for real work and bound head-of-line blocking (e.g., fixed-size cover frames).
- Evaluate congestion witnesses across hops (RTT time series, queue delay, loss/ECN) to avoid bottleneck shift.

1.4 Spectral surrogate evidence (why a “spectral tier” exists)

The PSC-Q main text treats a TV-tier contract and flags periodic/spectral tests as a realistic additional tier. Fig. 4 is an archive “spectral surrogate” plot illustrating that declared dither attenuates periodic witnesses, supporting the motivation for a two-tier view.

1.5 Interval-robust compilation against a utilization range

The “range compilation” note in the main text can be made more explicit: compile (δ, α) against a declared utilization interval $\rho \in [\rho_{\min}, \rho_{\max}]$ (rather than a point estimate), and size for $\rho_{\max}$ so the deployment remains within budget whenever the monitored regime stays within the published interval. Fig. 5 illustrates an archive sweep in which the chosen (δ, α) tightens as $\rho_{\max}$ increases.

References

- [1] Anonymous. PSC-Q: Public Service Calendars and Slack-Harvesting Substitution Padding for Congestion-Private Anonymous Overlays. Congestion series Paper 2, 2026. (This archive.)
- [2] Anonymous. Congestion-EQ: A Congestion Witness Contract for Anonymous Overlays. Congestion series Paper 1, 2026. (This archive.)

Figure omitted in source-only release. Requested file: `fig_service_bias_substitution.pdf`.

Figure 3: Archive microstudy: a service-bias pitfall when “cover” is implemented as competing work rather than strict idle-filling. Implementation takeaway: implement substitution as *preemptible* idle-filling (or as a separate low-priority queue that never delays real work).

Figure omitted in source-only release. Requested file: `fig_spectral_dither_attenuation.pdf`.

Figure 4: Archive evidence: dither attenuates periodic witnesses in a spectral surrogate.

- [3] Anonymous. Artifact policy and supporting materials for the anonymity/AnonDHT series. 2026.

Figure omitted in source-only release. Requested file: `fig_utilization_range_optimizer.pdf`.

Figure 5: Archive evidence: interval-robust compilation tightens as the assumed upper utilization bound increases.